

Institute of Actuaries of India

CT3: Probability and Mathematical Statistics

October 2009 Examination

Indicative Solutions

- Q. 1)** For 280 observations (which include the incorrect values of 64 and 80), sample mean = 54 and sample s.d. = 3.

Incorrect sample mean = $\sum \frac{x_i}{280} = 54$ where x_i is such that two of the values are incorrect.

Assume x_i^f is such that all the values are correct. Then

$$\text{Incorrect sample sum} = \sum x_i = 280 \times 54 = 15,120$$

$$\Rightarrow \text{correct sample sum } \sum x_i^f = 15,120 - (64 + 80) + (62 + 82) = 15,120$$

$$\Rightarrow \text{correct sample mean } \sum \frac{x_i^f}{280} = \frac{15,120}{280} = 54$$

$$\text{Again, incorrect sample s.d.} = \sqrt{\frac{1}{279} \{ \sum x_i^2 - 280 \bar{x}^2 \}} = 3$$

$$\Rightarrow \text{Incorrect sample sum of square} = \sum x_i^2 = 9 \times 279 + 280 \times \bar{x}^2 = 8,18,991$$

$$\Rightarrow \text{correct sample sum of square} = \sum x_i^{f2} = 8,18,991 - (64^2 + 80^2) + (62^2 + 82^2) = 8,19,063$$

$$\begin{aligned} \Rightarrow \text{correct sample s.d.} &= \sqrt{\frac{1}{279} \{ \sum x_i^{f2} - 280 \bar{x}^2 \}} = \sqrt{\frac{1}{279} \times \{ 8,19,063 - 280 \times 54^2 \}} \\ &= \sqrt{\frac{2388}{279}} = 3.0427 \end{aligned}$$

[3]

- Q. 2)** In a leap year (which consists of 366 days) there are 52 complete weeks and 2 days over.

The following are the possible combinations for these two “over” days:

- i. Sunday and Monday
- ii. Monday and Tuesday
- iii. Tuesday and Wednesday
- iv. Wednesday and Thursday
- v. Thursday and Friday
- vi. Friday and Saturday
- vii. Saturday and Sunday

In order that a leap year selected at random should contain 53 Sundays, one of the two “over” days must be Sunday.

Since out of the above 7 possibilities 2 are [(i) and (vii)] favourable to this event.

$$\text{Required probability} = \frac{2}{7}$$

[2]

Q. 3) X is a random variable, so total probability is equal to 1

$$\Rightarrow a + \int_0^1 b x^2 (1-x) dx = 1$$

$$\Rightarrow a + b \left(\int_0^1 x^2 dx - \int_0^1 x^3 dx \right) = 1$$

$$\Rightarrow a + b \left(\left[\frac{x^3}{3} \right]_0^1 - \left[\frac{x^4}{4} \right]_0^1 \right) = 1$$

$$\Rightarrow a + b \left(\frac{1}{3} - \frac{1}{4} \right) = 1$$

$$\Rightarrow a + \frac{b}{12} = 1 \dots\dots\dots(i)$$

$$E(X) = a \Rightarrow a \times 0 + \int_0^1 b x^3 (1-x) dx = a$$

$$\Rightarrow b \int_0^1 x^3 (1-x) dx = a$$

$$\Rightarrow b \left(\frac{1}{4} - \frac{1}{5} \right) = a$$

$$\Rightarrow b = 20a$$

$$\text{From (i), we get } a + \frac{20a}{12} = 1 \Rightarrow a = \frac{3}{8} \text{ or } 0.375 \text{ and } b = \frac{15}{2} \text{ or } 7.5$$

$$E(X^2) = \frac{3}{8} \times 0 + \frac{15}{2} \int_0^1 x^4 (1-x) dx \Rightarrow \frac{15}{2} \left(\frac{1}{5} - \frac{1}{6} \right) = \frac{1}{4}$$

$$S.D. \text{ of } X = \sqrt{\frac{1}{4} - \left(\frac{3}{8} \right)^2} = \frac{\sqrt{7}}{8} \text{ or } 0.3307$$

[3]

Q. 4) Assume X is discrete random variable, then $G(t) = \sum t^x P(X = x)$

$$\int_0^1 t^{(\alpha-1)} G(t) dt = \int_0^1 t^{(\alpha-1)} \sum t^x P(X = x) dt = \sum P(X = x) \int_0^1 t^{(\alpha-1)} t^x dt$$

$$= \sum P(X = x) \left[\frac{t^{x+\alpha}}{x+\alpha} \right]_0^1 = \sum \frac{1}{x+\alpha} P(X = x) = E\left(\frac{1}{x+\alpha}\right)$$

[2]

Q. 5) The marginal distribution of X is

$$f_1(x) = \int_{-1}^1 \left[\frac{1}{4} (1 + xy) \right] dy = \frac{1}{4} \left\{ [y]_{-1}^1 + x \left[\frac{y^2}{2} \right]_{-1}^1 \right\} = \frac{1}{2}, -1 \leq x \leq 1.$$

$$\text{Similarly the marginal distribution of Y is } f_1(y) = \frac{1}{2}, -1 \leq y \leq 1.$$

$$\text{Now, } f_1(x) \times f_1(y) \neq f(x, y)$$

Hence X and Y are not independent.

$$P(X^2 \leq x) = P(|X| \leq \sqrt{x}) = \int_{-\sqrt{x}}^{\sqrt{x}} f_1(u) du = \left[\frac{u}{2} \right]_{-\sqrt{x}}^{\sqrt{x}} = \sqrt{x}$$

$$\text{Similarly, } P(Y^2 \leq y) = \sqrt{y}$$

$$\begin{aligned} P(X^2 \leq x \cap Y^2 \leq y) &= P(|X| \leq \sqrt{x} \cap |Y| \leq \sqrt{y}) = \int_{-\sqrt{x}}^{\sqrt{x}} \left[\int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{4} (1+uv) du \right] dv \\ &= \int_{-\sqrt{x}}^{\sqrt{x}} \left[\frac{1}{4} \left\{ [u]_{-\sqrt{y}}^{\sqrt{y}} + v \left[\frac{u^2}{2} \right]_{-\sqrt{y}}^{\sqrt{y}} \right\} \right] dv = \int_{-\sqrt{x}}^{\sqrt{x}} \frac{\sqrt{y}}{2} dv = \frac{\sqrt{y}}{2} [v]_{-\sqrt{x}}^{\sqrt{x}} = \frac{\sqrt{y}}{2} \times 2\sqrt{x} = \sqrt{x} \times \sqrt{y} \\ &= P(X^2 \leq x) \times P(Y^2 \leq y) \end{aligned}$$

Hence X^2 and Y^2 are independent.

[5]

Q. 6)

$$X \sim P(\lambda) \Rightarrow p_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$\begin{aligned} E|X-1| &= \sum_{x=0}^{\infty} |x-1| \frac{\lambda^x e^{-\lambda}}{x!} \\ &= \sum_{x=0}^{\infty} |x-1| \frac{\lambda^x e^{-\lambda}}{x!} + \sum_{x=1}^{\infty} |x-1| \frac{\lambda^x e^{-\lambda}}{x!} + \sum_{x=2}^{\infty} |x-1| \frac{\lambda^x e^{-\lambda}}{x!} \\ &= e^{-\lambda} + 0 + \sum_{x=2}^{\infty} |x-1| \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} + \sum_{x=2}^{\infty} (x-1) \frac{\lambda^x e^{-\lambda}}{x!} \\ &= e^{-\lambda} + e^{-\lambda} \sum_{x=2}^{\infty} (x-1) \frac{\lambda^x}{x!} = e^{-\lambda} + e^{-\lambda} \left[\sum_{x=2}^{\infty} \frac{\lambda^x}{(x-1)!} - \sum_{x=2}^{\infty} \frac{\lambda^x}{x!} \right] \\ &= e^{-\lambda} + \left[\lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1} e^{-\lambda}}{(x-1)!} - \sum_{x=2}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} \right] \\ &= e^{-\lambda} + [\lambda (1 - e^{-\lambda}) - (1 - e^{-\lambda} - \lambda e^{-\lambda})] \\ &= e^{-\lambda} + [\lambda - \lambda e^{-\lambda} - 1 + e^{-\lambda} + \lambda e^{-\lambda}] \\ &= e^{-\lambda} + \lambda - 1 + e^{-\lambda} \\ &= \lambda - 1 + 2e^{-\lambda} \end{aligned}$$

[4]

Q. 7) As X is a random variable with probability function $p(x)$,

(a)

$$\sum_{x=1}^{\infty} p(x) = 1 \Rightarrow \sum_{x=1}^{\infty} \frac{c}{x(x+1)} = 1 \Rightarrow c \sum_{x=1}^{\infty} \left[\frac{1}{x} - \frac{1}{x+1} \right] = 1$$

$$\Rightarrow C \left[\left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots \dots \dots \right] = 1 \Rightarrow C = 1$$

- (b) $E(X) = \sum_{x=1}^{\infty} x p(x) = \sum_{x=1}^{\infty} \frac{x}{x(x+1)} = \sum_{x=1}^{\infty} \frac{1}{(x+1)}$ which is a divergent series and hence $E(X)$ does not exist.

- (c) Moment Generating Function $M_X(t) = E(e^{tx}) = \sum_{x=1}^{\infty} e^{tx} p(x) = \sum_{x=1}^{\infty} e^{tx} \frac{1}{x(x+1)}$
- $$= \sum_{x=1}^{\infty} \frac{e^{tx}}{x(x+1)}; \text{ where } z = e^t$$
- $$= \frac{z}{1 \cdot 2} + \frac{z^2}{2 \cdot 3} + \frac{z^3}{3 \cdot 4} + \frac{z^4}{4 \cdot 5} + \dots \dots \dots$$
- $$= z \left(1 - \frac{1}{2} \right) + z^2 \left(\frac{1}{2} - \frac{1}{3} \right) + z^3 \left(\frac{1}{3} - \frac{1}{4} \right) + z^4 \left(\frac{1}{4} - \frac{1}{5} \right) + \dots \dots \dots$$
- $$= \left[z + \frac{z^2}{2} + \frac{z^3}{3} + \frac{z^4}{4} + \dots \dots \dots \right] - \left[\frac{z}{2} + \frac{z^2}{3} + \frac{z^3}{4} + \frac{z^4}{5} + \dots \dots \dots \right]$$
- $$= -\log(1-z) - \frac{1}{z} \left[\frac{z^2}{2} + \frac{z^3}{3} + \frac{z^4}{4} + \frac{z^5}{5} + \dots \dots \dots \right]$$
- $$= -\log(1-z) - \frac{1}{z} [-\log(1-z) - z]$$
- $$= -\log(1-z) + \frac{1}{z} \log(1-z) + 1$$
- $$= 1 + \left(\frac{1}{z} - 1 \right) \log(1-z)$$
- $$= 1 + \log(1-z)^{\left(\frac{1}{z} - 1 \right)}$$
- $$\Rightarrow M_X(t) - 1 = \log(1-z)^{\left(\frac{1}{z} - 1 \right)}$$
- $$\Rightarrow (1-z)^{\left(\frac{1}{z} - 1 \right)} = e^{(M_X(t) - 1)}$$
- $$\Rightarrow (1 - e^t)^{\left(\frac{1 - e^t}{e^t} \right)} = e^{(M_X(t) - 1)}$$

[6]

Q.8) The basic model would be:

$$E(Y | x_1, x_2) = \alpha + \beta_1 x_1 + \beta_2 x_2$$

Where,

x_1 = number of actuarial papers cleared

x_2 = number of years of professional experience in life insurance industry

Y = Salary

α = (a constant), the average salary of new actuarial staff (with no paper cleared or professional experience)

β_1 = (a constant), the changes in salary associated with actuarial papers clearing

β_2 = (a constant), the changes in salary associated with years of professional experience in life insurance.

Since the data relates to 27 (= n) actuarial staffs, we need to introduce a sub script i corresponding to i^{th} staff.

So the expression of the model for actual salary for the i^{th} actuarial staff will be:

$$Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \epsilon_i$$

Where, ϵ_i is the difference between the actuarial staff's actual salary and the theoretical salary for someone with the same number of actuarial papers cleared and years of professional experience in life insurance industry

[Candidates give the expression of the model without mentioning $E(Y/X_1, X_2)$ may be given full credit]

[3]

- Q.9)
- (a) $F_U(u) = P(U \leq u) = 1 - P(U > u) = 1 - [P(X_i > u)]^n = 1 - \left[\int_u^1 dx \right]^n = 1 - (1-u)^n$
 $f_U(u) = n(1-u)^{n-1}, \quad 0 \leq u \leq 1$
- (b) $E(U) = \int_0^1 n u (1-u)^{n-1} du = -n \left[u \frac{(1-u)^n}{n} \right]_0^1 + \int_0^1 (1-u)^n du = \left[-\frac{(1-u)^{n+1}}{n+1} \right]_0^1 = \frac{1}{n+1}$
 $E(U^2) = \int_0^1 n u^2 (1-u)^{n-1} du = -n \left[u^2 \frac{(1-u)^n}{n} \right]_0^1 + \int_0^1 2u (1-u)^n du = \frac{2}{(n+2)(n+1)}$
 $V(U) = \frac{2}{(n+1)(n+2)} - \left(\frac{1}{n+1} \right)^2 = \frac{n}{(n+2)(n+1)^2}$
 $F_V(v) = P(V \leq v) = [P(X_i \leq v)]^n = \left[\int_0^v dx \right]^n = v^n$
 $f_V(v) = n v^{n-1}, \quad 0 \leq v \leq 1$
- (c) $E(V) = \int_0^1 n v v^{n-1} dv = n \left[\frac{v^{n+1}}{n+1} \right]_0^1 = \frac{n}{n+1} = n E(U)$
 $E(V^2) = \int_0^1 n v^2 v^{n-1} dv = n \left[\frac{v^{n+2}}{n+2} \right]_0^1 = \frac{n}{n+2}$
 $V(V) = \frac{n}{(n+2)} - \left(\frac{n}{n+1} \right)^2 = \frac{n}{(n+2)(n+1)^2} = V(U)$

[8]

- Q.10) Critical region is the region of rejection of H_0 when the value of test statistic falls in this region.

(a)

Here $H_0: p = 0.5$ and $H_1: p = 0.6$

(b)

If the random variable X denotes the number of heads in 10 trials of the coin then

$$X \sim \text{Bin}(10, p)$$

α = size of the test = $P(\text{reject } H_0 \text{ when } H_0 \text{ is true}) = P(X \geq 7 \mid p = 0.5)$

$$= \sum_{x=7}^{10} \binom{10}{x} (0.5)^x (1-0.5)^{10-x} = 0.1719$$

Power of the test = $P(\text{reject } H_0 \text{ when } H_0 \text{ is false}) = P(X \geq 7 \mid p = 0.6)$

$$= \sum_{x=7}^{10} \binom{10}{x} (0.6)^x (1-0.6)^{10-x} = 0.3823$$

[5]

Q.11) Given $X \sim N(\mu, 1)$ and $Y \sim N(\rho\mu, 1)$ independently.

(a)

Since the **linear combination** of two normally distributed random variables are themselves normally distributed, obtain $Z \sim N(E(Z), \text{Var}(Z))$, where $Z = Y - \rho X$

$$E(Z) = E(Y - \rho X) = E(Y) - \rho E(X) = \rho\mu - \rho\mu = 0$$

$$\text{Var}(Z) = \text{Var}(Y - \rho X) = \text{Var}(Y) + \rho^2 \text{Var}(X), \text{ as } X \text{ and } Y \text{ are independent}$$

$$= 1 + \rho^2$$

Hence $Z = Y - \rho X \sim N(0, 1 + \rho^2)$, which depends on ρ , but not on μ .

(b) $H_0: \rho = 1$ $H_1: \rho \neq 1$

Under H_0 , $Z = Y - \rho X = Y - X \sim N(0, 2)$

Hence, $\frac{1}{\sqrt{2}} (Y - X) \sim N(0, 1)$

$$\text{Observed value of } \frac{1}{\sqrt{2}} (Y - X) = \frac{1}{\sqrt{2}} \times 2.4 = 1.697$$

As observed value $1.697 < z_{0.05} = 1.96$, we accept H_0

If $\rho = 1$, $E(Y) = E(X)$

(c) Despite of observed ratio $\frac{Y}{X} = \frac{2.4}{0.3} = 9$, i.e. $y = 9x$, $H_0: \rho = 1$ is accepted

[Candidates give the equivalent comments may be given full credit]

[5]

Q.12) Let $S = \sum_{i=1}^n (Y_i - E(Y_i))^2$ has to be minimized,

(a)

$$S = \sum_{i=1}^n (Y_i - E(Y_i))^2 = \sum_{i=1}^n (Y_i - \theta x_i)^2$$

$$\frac{dS}{d\theta} = -2 \sum_{i=1}^n x_i (Y_i - \theta x_i), \text{ Setting } \frac{dS}{d\theta} = 0, \text{ we get } \sum_{i=1}^n x_i Y_i - \theta \sum_{i=1}^n x_i^2 = 0$$

$$\Rightarrow \bar{\theta} = \frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{d^2 S}{d\theta^2} = 2 \sum_{i=1}^n x_i^2 > 0$$

Hence, $\bar{\theta}$ is the least square estimator

(b) $L(\theta) = e^{-\sum x_i \theta} \prod (x_i \theta)^{x_i} \times \text{Constant}$

$$\log(L(\theta)) = -\theta \sum x_i + \sum Y_i \log(\theta) + \text{Constant}$$

$$\frac{d \log(L(\theta))}{d\theta} = -\sum x_i + \frac{1}{\theta} \sum Y_i, \text{ Setting to } 0 \Rightarrow \bar{\theta} = \frac{\sum Y_i}{\sum x_i}$$

$$\frac{d^2 \log(L(\theta))}{d\theta^2} = -\frac{1}{\theta^2} \sum Y_i = \left[-\frac{1}{\theta^2} \sum Y_i \right]_{\theta=\bar{\theta}} = -\frac{\sum Y_i}{\bar{\theta}^2} < 0$$

Hence, $\bar{\theta}$ is the maximum likelihood estimator

$$(c) \quad E(\bar{\theta}) = E\left(\frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2}\right) = \frac{1}{\sum_{i=1}^n x_i^2} E\left(\sum_{i=1}^n x_i Y_i\right) = \frac{1}{\sum_{i=1}^n x_i^2} \left(\sum_{i=1}^n x_i \theta x_i\right) = \theta$$

Hence unbiased estimator

$$E(\bar{\theta}) = E\left(\frac{\sum_{i=1}^n Y_i}{\sum_{i=1}^n x_i}\right) = \frac{1}{\sum_{i=1}^n x_i} E\left(\sum_{i=1}^n Y_i\right) = \frac{1}{\sum_{i=1}^n x_i} \left(\sum_{i=1}^n \theta x_i\right) = \theta$$

Hence unbiased estimator

$$(d) \quad \text{From the given data,}$$

$$\sum x_i = 182,166; \quad \sum y_i = 1,799; \quad \sum x_i y_i = 106,410,416; \quad \sum x_i^2 = 9,220,812,676$$

$$\Rightarrow \bar{\theta} = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{106,410,416}{9,220,812,676} = 0.01154$$

$$\Rightarrow \bar{\theta} = \frac{\sum y_i}{\sum x_i} = \frac{1,799}{152,166} = 0.00988$$

[11]

Q.13) (i) To test $H_0: \mu_X = \mu_Y$ against $H_1: \mu_X \neq \mu_Y$

(a)

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_1} + \frac{\sigma_Y^2}{n_2}}} \sim N(0,1)$$

$$\bar{X} = 92.255; \quad \bar{Y} = 92.753; \quad n_1 = n_2 = 8; \quad \sigma_X^2 = 8.04; \quad \text{and} \quad \sigma_Y^2 = 14.84$$

$$Z_{\text{observed}} = -0.282; \quad Z_{0.05} = 1.96;$$

Accept H_0

(ii) Variances are equal and unknown, so combined variance $s_p^2 = \frac{(n_1-1)s_X^2 + (n_2-1)s_Y^2}{n_1 + n_2 - 2}$

$$s_X^2 = 5.688; \quad s_Y^2 = 8.901; \quad s_p^2 = 7.295$$

$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t \text{ distribution with d.f. } (n_1 + n_2 - 2)$$

$$t_{\text{observed}} = -0.354; \quad t_{0.05, 14} = 2.145;$$

Accept H_0

(b) To test $H_0: \sigma_X^2 = \sigma_Y^2$ against $H_1: \sigma_X^2 \neq \sigma_Y^2$

$$F = \frac{s_X^2/(n_1-1)}{s_Y^2/(n_2-1)} \sim F \text{ distribution with d.f. } (n_1 - 1, n_2 - 1)$$

$$F_{\text{observed}} = 0.639; \quad F_{(5, 7), 0.025} = 4.99; \quad F_{(5, 7), 0.975} = 1/4.99 = 0.2004;$$

Accept H_0

- (c) Confidence intervals of the difference of mean yields i.e. $(\mu_x - \mu_y)$ at 95% level under equal of variance is given by,

$$(\bar{x} - \bar{y}) \pm t_{0.025, 8} s_p \sqrt{\frac{1}{n_x} + \frac{1}{n_y}}$$

$$(92.255 - 92.733) \pm 2.145 \times 2.701 \times \sqrt{\frac{1}{8} + \frac{1}{8}} = (-3.574, 2.419)$$

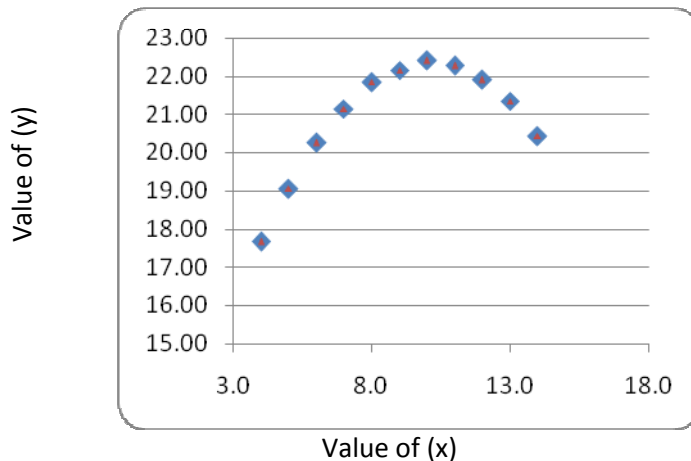
- (d) Since the confidence interval includes 0, we accept the hypothesis $H_0: \mu_x = \mu_y$, as already tested above.

[14]

Q.14)

(a)

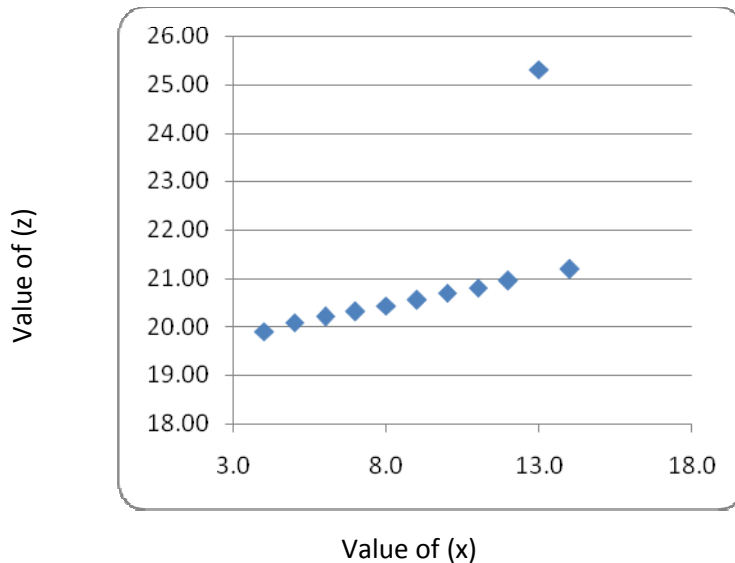
Scatter plot of y against x



From the scatter plot we see that the line of regression may not be a best fit, but a polynomial regression equation may fit.

b)

Scatter plot of z against x



From the scatter plot we see that the line of regression may be a best fit. Only a point seems to be an outlier.

(c) $\bar{x} = 9.00$; $\bar{y} = 20.95$; $n = 11$;

$$S_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = 1,001.00 - 11 \times (9.00)^2 = 110.00$$

$$S_{yy} = \sum (y_i - \bar{y})^2 = \sum y_i^2 - n\bar{y}^2 = 4,850.30 - 11 \times (20.95)^2 = 22.37$$

$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y} = 2,104.85 - 11 \times (9.00 \times 20.95) = 30.80$$

$$\hat{\beta}_{xy} = \frac{S_{xy}}{S_{xx}} = \frac{30.80}{110.00} = 0.28$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}_{xy}\bar{x} = 20.95 - 0.28 \times 9.00 = 18.43$$

The least squares fitted regression line of y on x = $18.43 + 0.28x$

(d) $\sum z = 230.45$; $\bar{z} = 20.95$; $\sum z^2 = 4,850.30$; $\sum xz = 2,104.85$

$$S_{zz} = \sum (z_i - \bar{z})^2 = \sum z_i^2 - n\bar{z}^2 = 4,850.30 - 11 \times (20.95)^2 = 22.37$$

$$S_{xz} = \sum (x_i - \bar{x})(z_i - \bar{z}) = \sum x_i z_i - n\bar{x}\bar{z} = 2,104.85 - 11 \times (9.00 \times 20.95) = 30.80$$

So the least square fitted regression line of z on x is same as obtained in (c) above = $18.43 + 0.28x$

(e) Correlation coefficient of x and y (ρ_{xy}) = $\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{30.80}{\sqrt{110.00 \times 22.37}} = 0.621$

Also (ρ_{xz}) = $\frac{S_{xz}}{\sqrt{S_{xx}S_{zz}}} = \frac{30.80}{\sqrt{110.00 \times 22.37}} = 0.621$

Though the relationship of y on x and z on x are different, as found in the above scatter plots, the correlation coefficient are same and positive.

(f)(i) After omitting the values for city 3, we calculate,

$$\bar{x} = 8.60; \quad \bar{z} = 20.51; \quad n = 10; \quad \sum x^2 = 832.00, \quad \sum z^2 = 4,209.71, \quad \sum xz = 1,775.82$$

$$S_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = 832.00 - 10 \times (8.60)^2 = 92.40$$

$$S_{zz} = \sum (z_i - \bar{z})^2 = \sum z_i^2 - n\bar{z}^2 = 4,209.71 - 10 \times (20.51)^2 = 1.46$$

$$S_{xz} = \sum (x_i - \bar{x})(z_i - \bar{z}) = \sum x_i z_i - n\bar{x}\bar{z} = 1,775.82 - 10 \times (8.60 \times 20.51) = 11.62$$

$$\hat{\beta}_{xz} = \frac{S_{xz}}{S_{xx}} = \frac{11.62}{92.40} = 0.126$$

$$\hat{\alpha} = \bar{z} - \hat{\beta}_{xz}\bar{x} = 20.51 - 0.126 \times 8.60 = 19.43$$

The regression line of z on x = $19.43 + 0.126x$

(ii) (ρ_{xz}) = $\frac{S_{xz}}{\sqrt{S_{xx}S_{zz}}} = \frac{11.62}{\sqrt{92.40 \times 1.46}} = 0.999$

(g) After omitting the outlier value for city 3, the correlation coefficient is approximately equal to 1, i.e. the minimum temperatures on a certain day is perfectly correlated with the maximum temperatures on weekend on the basis of the data provided.

Q.15)

(a) The mathematical model of one-way ANOVA is given by

$$Y_{ij} = \mu + \tau_i + e_{ij}; \quad i = 1, 2, \dots, k \quad j = 1, 2, \dots, n_i$$

where,

k = number of treatments

 n_i = number of responses from i^{th} treatment Y_{ij} is the j^{th} response from i^{th} treatment τ_i is the i^{th} treatment μ is the over mean response e_{ij} is the error termAssumption : e_{ij} is i.i.d $N(0, \sigma^2)$

(b)

 H_0 : Mean claim amounts of five companies are equal H_1 : Mean claim amounts of five companies are not equalWe have $n_1 = 7, n_2 = 8, n_3 = 6, n_4 = 7, n_5 = 5, n = 33$

$$\sum_{i=1}^5 \sum_{j=1}^{n_i} y_{ij} = 1,856$$

$$\sum_{i=1}^5 \sum_{j=1}^{n_i} y_{ij}^2 = 174,316$$

$$C.F. = \left(\sum_{i=1}^5 \sum_{j=1}^{n_i} y_{ij} \right)^2 / n = 104,385.94$$

$$SS_T = 174316 - C.F. = 69,930.06;$$

$$SS_B = \sum_{i=1}^5 \left(\sum_{j=1}^{n_i} y_{ij} \right)^2 / n_i - C.F. = 354^2 / 7 + 386^2 / 8 + 87^2 / 6 + 645^2 / 7 + 384^2 / 5 - 104,385.94$$

$$= 22,325.69$$

$$SS_R = SS_T - SS_B = 47,604.37$$

<u>Sources of Variation</u>	<u>d.f</u>	<u>SS</u>	<u>MSS</u>	<u>F</u>
Companies	4	22,326	5,581.42	3.283
Residual	28	47,604	1,700.16	
Total	32	69,930		

$$F_{\text{observed}} = 3.283; \quad F_{(4, 28, 0.05)} = 2.714;$$

Reject H_0

(c) H_0 : Salaries are independent of number of actuarial papers cleared

H_1 : Salaries are dependent on number of actuarial papers cleared

Observed Values (O_i)

Papers cleared	Salary per annum (in Rs. lacs)				Total
	3 - 5	5 - 8	8 - 10	10 - 12	
0 - 3	45	20	6	5	76
4 - 6	7	20	9	6	42
7 - 9	5	8	15	12	40
Total	57	48	30	23	158

Under H_0 , Expected Values (E_i)

Papers cleared	Salary per annum (in Rs. lacs)				Total
	3 - 5	5 - 8	8 - 10	10 - 12	
0 - 3	27.42	23.09	14.43	11.06	76.00
4 - 6	15.15	12.76	7.97	6.11	42.00
7 - 9	14.43	12.15	7.59	5.82	40.00
Total	57.00	48.00	30.00	23.00	158.00

$$\chi^2 = \sum_{i=1}^{12} (O_i - E_i)^2 / E_i = (27.42 - 45)^2 / 27.42 + \dots + (5.82 - 12)^2 / 5.82 = 49.919$$

$$\chi^2_{\text{observed}} = 49.91 \quad \chi^2_{(6, 0.05)} = 12.59, \text{ where d.f. } 6 = (3-1) \times (4-1)$$

Reject H_0

[10]

Total Marks [100]
